Midterm (practice version - NOT GRADED), Math 170S, Winter 2018
Instructor: Elizaveta Rebrova

Printed name: ____________________

Signed name: ____________________

Student ID number: ____________________

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).

   (a) (5 points) Let $X_1, X_2, \ldots, X_{20}$ are samples taken from normal $N(\mu_X, 2)$ distribution $(2 = \sigma^2)$, $Y_1, Y_2, \ldots, Y_{40}$ are samples taken from normal $N(\mu_Y, 1)$ distribution, $Z_1, Z_2, \ldots, Z_{45}$ are taken from normal $N(\mu_Z, 5)$ distribution. All $X_i's$, $Y_i's$ and $Z_i's$ are mutually independent. What is the distribution of a random variable $\overline{X} - 2\overline{Y} + 3\overline{Z}$?

   (b) (5 points) Find 90% confidence interval for a parameter $\mu_X - 2\mu_Y + 3\mu_Z$.

2. (10 points) We have 2 boxes, each containing 3 balls. Box number 1 contains one black and two white balls; box number 2 contains two black and one white ball.

   Our friend chooses one of the boxes at random, probability of choosing box number 1 is p. Then he takes one ball from a chosen box (each of three balls can be taken chosen equally likely), and it turns out to be white.

   We are going to find MAP estimate for the parameter $\theta = 1$ or 2 to conclude which box was chosen (θ is the number of the box). For what values of p we can conclude that box number 1 was chosen? For what values of p we can conclude that box number 2 was chosen?

3. Consider a sequence of independent coin tosses, and let θ be the probability of heads at each toss.

   (a) (4 points) Let k be a fixed number, and N is the number of tosses until k-th head occurs. Find maximum likelihood estimator of θ (depending of N).

   (b) (4 points) Now let n be a fixed number, and K is the number of heads in n tosses. Find maximum likelihood estimator of θ (depending on K).

   (c) (2 points) Is the estimator from part (b) biased?

   (d) (points) (Optional) Is the estimator from part (a) biased?

4. (a) (4 points) Let $X_1, X_2, \ldots, X_n$ are samples taken from the uniform distribution on $[\theta, 1]$ $(\theta < 1)$. Show that $X_{(1)}$ is a sufficient statistic for θ.

(b) (6 points) Let $X_1, X_2, \ldots, X_n$ are samples taken from the uniform distribution on $[\theta, \theta + 1]$. What is the length of a confidence interval with confidence coefficient $1 - \alpha$ for the estimate $X_{(1)}$ for parameter θ?

5. (10 points) Let $X_1, X_2, \ldots, X_n$ be samples taken from some distribution with unknown density function. However, you know that $\mathbb{E}X = \alpha/\beta$ and $\operatorname{Var} X = \alpha/\beta^2$, where α and β are two parameters of the density function (and there is no third parameter). What method could we use to find good estimates for α and β? What are these estimates?